

Convex Sets and Convex Functions

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1 Introduction

The notion of convexity appears to play an important role in many branches of Mathematics and Physics such as Analysis, Geometry, Optimization, Mechanics, etc,. In these notes we present the properties of convex sets and convex functions.

2 Convex sets

Definition 2.1 (Convex set)

Let V be a vector space over the scalar field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. A set $C \subset V$ is said to be convex if either C is a singleton or the segment line

$$[x, y] := \{\lambda x + (1 - \lambda)y : \lambda \in [0, 1]\}$$

joining any two points x and y of C lies entirely in C .

Example 2.2 (Examples of Convex sets)

- (a) Given two vectors x and v in $\mathbb{R}^n$. The ray (or half-line) H from x in the direction of v defined by

$$H := \{x + \lambda v : \lambda \geq 0\}$$

is a convex subset of $\mathbb{R}^n$.

- (b) Linear subspaces of the vector space V are convex sets.

- (c) Given $x^* \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. The *closed half-spaces*

$$F^+ := \{y \in \mathbb{R}^n : x^* \cdot y \geq \alpha\} \quad \text{and} \quad F^- := \{y \in \mathbb{R}^n : x^* \cdot y \leq \alpha\},$$

and the *open half-spaces*

$$G^+ := \{y \in \mathbb{R}^n : x^* \cdot y > \alpha\} \quad \text{and} \quad G^- := \{y \in \mathbb{R}^n : x^* \cdot y < \alpha\}$$

determined by x^* and α are all convex sets.

- (d) Let $(V, \|\cdot\|_V)$ be a normed space. Given $x \in V$ and $r > 0$. Then the open and closed balls $B(x, r)$ and $B[x, r]$ centered at x with radius r are convex sets. For instance, to show that $B[x, r]$ is convex, we take $y, z \in B[x, r]$, $\lambda \in [0, 1]$ and we want to show that $\lambda y + (1 - \lambda)z \in B[x, r]$. That is,

$$\|\lambda y + (1 - \lambda)z - x\|_V \leq r? \quad \text{(Check!)}$$

Hint: $x = \lambda x + (1 - \lambda)y$.

- (e) Given an $n \times m$ matrix $A = (a_{ij})$ and a vector $b \in \mathbb{R}^n$. The set of solutions $x \in \mathbb{R}^m$ of the system of linear inequalities

[illegible]

denoted by $Ax < b$ is also a convex subset of $\mathbb{R}^n$. **Check!**

- (f) Let $x^{(1)}, x^{(2)}, \dots, x^{(k)}$ be vectors in $\mathbb{R}^n$. We call *convex combination* of $x^{(1)}, x^{(2)}, \dots, x^{(k)}$ any vector

$$y := \lambda_1 x^{(1)} + \lambda_2 x^{(2)} + \cdots + \lambda_k x^{(k)} = \sum_{i=1}^k \lambda_i x^{(i)} \quad \text{with } \lambda_1, \lambda_2, \dots, \lambda_k \geq 0 \text{ such that } \sum_{i=1}^k \lambda_i = 1.$$

The set $\text{Conv}(x^{(1)}, \dots, x^{(k)})$ of convex combinations of the vectors $x^{(1)}, x^{(2)}, \dots, x^{(k)}$ is a convex set in $\mathbb{R}^n$. **Check!**

3 Convex functions

Definition 3.1 (Convex function)

Let V be a vector space over the scalar field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. Let C be a convex subset of V . We say that the function $f : C \rightarrow \mathbb{R}$ is

- (i) *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad \forall x, y \in C \quad \forall \lambda \in [0, 1]; \quad (1)$$

- (ii) *strictly convex* if we have a strict inequality in (1) for all $x, y \in C$ with $x \neq y$ and all $\lambda \in (0, 1)$.

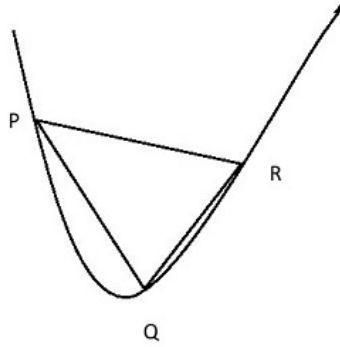
We will say that f is *concave* (resp. *strictly concave*) on C if $-f$ is convex (resp. strictly convex).

Exercise 3.2 Using mathematical induction on k , prove that f is convex on the convex set C if for every k positive integer, for all nonnegative numbers $\lambda_1, \lambda_2, \dots, \lambda_k$ whose sum is 1 and all vectors $x^{(1)}, x^{(2)}, \dots, x^{(k)} \in C$ we have

$$\boxed{f\left(\sum_{i=1}^k \lambda_i x^{(i)}\right) \leq \sum_{i=1}^k \lambda_i f(x^{(i)})}.$$

3.1 Single variable convex functions

- (a) Geometrically, the inequality (1) means that if P , Q , and R are any three points on the graph of f with Q between P and R (**the order!**), then Q is on or below the chord PR (see Fig. above).



In terms of slope, it is equivalent to

$$\text{slope } PQ \leq \text{slope } PR \leq \text{slope } QR \quad (2)$$

with strict inequalities when f is strictly convex.

This description of convex function is used to establish the continuity of single variable convex functions

- (b) If f is a convex function defined on a closed interval $[a, b]$, then f is continuous on (a, b) .

Proof: Let $x \in (a, b)$. We want to prove that f is continuous at x . We prove that f is continuous at the right of x . Fix two numbers s and t in (a, b) such that $s < x < t$. Now let y be a number varying in the interval (x, t) . Hence we have

$$s < x < y < t.$$

From the geometric interpretation of convexity in (2), we have that the point $(y, f(y))$ lies on or above the line through $(s, f(s))$ and $(x, f(x))$, but lies on or below the line through $(x, f(x))$ and $(t, f(t))$. That is,

$$f(s) + \frac{f(x) - f(s)}{x - s} (y - s) \leq f(y) \leq f(x) + \frac{f(t) - f(x)}{t - x} (y - x). \quad (3)$$

Therefore, by Sandwich theorem we find that

$$\lim_{y \rightarrow x^+} f(y) = f(x).$$

Using similar arguments we find that $\lim_{z \rightarrow x^-} f(z) = f(x)$.

Hence, f is continuous at x as being continuous from the right and from the left. ■

- (c) It may happen to have discontinuities at the end-points of the interval. In fact the function $f(x) = x^2$ if $x \in (-1, 1)$ and $f(-1) = f(1) = 2$ is convex on $[-1, 1]$ but it is not continuous at the endpoints of its domain as it has upward jumps there.
- (d) It is easy that f is bounded from above. In fact, set $M = \max(f(a), f(b))$. for every $x \in [a, b]$, $x = \lambda a + (1 - \lambda)b$ for some $\lambda \in [0, 1]$. Hence

$$\boxed{f(x) \leq \lambda f(a) + (1 - \lambda)f(b) \leq \lambda M + (1 - \lambda)M = M} \quad (*)$$

We also have that f is bounded from below. In fact, any arbitrary point $x \in [a, b]$ can be written in the form $x = \frac{a+b}{2} + t$. since

$$\frac{a+b}{2} = \frac{1}{2} \left[\frac{a+b}{2} + t \right] + \frac{1}{2} \left[\frac{a+b}{2} - t \right],$$

with $\frac{a+b}{2} - t = a + b - x$ and hence $\frac{a+b}{2} - t \in [a, b]$, we obtain that

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{2}f\left(\frac{a+b}{2} + t\right) + \frac{1}{2}f\left(\frac{a+b}{2} - t\right).$$

Hence

$$\boxed{f(x) = f\left(\frac{a+b}{2} + t\right) \geq 2f\left(\frac{a+b}{2}\right) - f\left(\frac{a+b}{2} - t\right) \geq 2f\left(\frac{a+b}{2}\right) - M = m} \quad (**)$$

Therefore, it follows from (*) and (**) that

$$m \leq f(x) \leq M \quad \text{for every } x \in [a, b]$$

with $M = \max(f(a), f(b))$ and $m = 2f\left(\frac{a+b}{2}\right) - M$. □

3.1.1 Examples of convex functions

- (a) $f(x) = |x|$ is convex on $\mathbb{R}$.
- (b) Any affine function f on $\mathbb{R}^n$ is convex. That is, fix $a \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

$$f(\mathbf{x}) = a \cdot \mathbf{x} + \alpha \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

- (c) Given a vector $a \in \mathbb{R}^n$, the function $f(x) = (a \cdot x)^2$, $x \in \mathbb{R}^n$ is convex.
- (d) In general, let f be a convex function on $\mathbb{R}$. Let a be a fixed vector on $\mathbb{R}^n$ and let α be a fixed real number. Then the function g defined on $\mathbb{R}^n$ by $g(x) = f(a \cdot x + \alpha)$ is convex.

The following proposition shows the interest of convex function in optimization

Proposition 3.3 *Let f be convex function defined on the convex set C . Then any minimizer of f is also a global minimizer. If f is strictly convex on then convex set C , then any local minimizer is the unique global minimizer of f on C .*

Proof: let x_0 be a local minimizer for the convex function f on C . There exist a positive number r_0 such that

$$f(x_0) \leq f(x) \quad \text{for all } x \in B(x_0, r_0) \cap C.$$

Let $y \in C$ we want to show that $f(x_0) \leq f(y)$. Let $\lambda \in (0, 1)$ be such that

$$x_0 + \lambda(y - x_0) \in C \quad \text{and} \quad \|x_0 + \lambda(y - x_0) - x_0\| < r.$$

To find such a λ , we just consider the segment $[x_0, y]$.

Then we get

$$f(x_0) \leq f(x_0 + \lambda(y - x_0)) = f(\lambda y + (1 - \lambda)x_0) \leq \lambda f(y) + (1 - \lambda)f(x_0).$$

Now subtracting $f(x_0)$ from both sides and dividing by λ we get that $f(x_0) \leq f(y)$.

If f is strictly convex, we get similarly that x_0 a global minimizer of f on the convex set C , that is $f(x_0) = m$, the minimum value of f on C . Let us prove that x_0 is the unique strict global minimizers. Assume by contradiction that f has another strict global minimizer y_0 on C with $x_0 \neq y_0$. Then by strict convexity of f we have

$$f\left(\frac{1}{2}x_0 + \frac{1}{2}y_0\right) < \frac{1}{2}f(x_0) + \frac{1}{2}f(y_0) = \frac{m}{2} + \frac{m}{2} = m.$$

Therefore we have found another vector $z_0 = \frac{1}{2}x_0 + \frac{1}{2}y_0 \in C$ (because C is convex and $x_0, y_0 \in C$) such that $f(z_0) < m$ which is impossible since m is the minimum value of f on C . ■

The next lecture will be on differentiable convex functions.